Prochoice Analysis

Load the data

```r
library(readxl)

file_path <- "/Users/yaldasmac/Desktop/prochoice_intervention.xlsx"

data <- read_excel(file_path)
```

Pepare the data

```r
library(dplyr)
```

```
Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

    filter, lag

The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

```r
# Create mean scores for your key variables
data <- data %>%
  mutate(
    threat_score = rowMeans(select(., threat_1:threat_6), na.rm = TRUE),
    trust_score = rowMeans(select(., trust_1:trust_6), na.rm = TRUE),
    perspective_taking = rowMeans(select(., emp_1:emp_14), na.rm = TRUE)
  )

# Convert intervention to a factor with meaningful labels
data <- data %>%
  mutate(
    intervention_factor = factor(intervention, levels = c(1, 2, 3),
                                labels = c("Better", "Worse", "Control")),
    group_factor = factor(group, levels = c(1, 2),
                         labels = c("Individualizing", "Binding"))
  )
```

Run the ANCOVA Models with Moderation

```r
# For trust outcome
trust_model <- aov(trust_score ~ ideology + intervention_factor * group_factor,
                  data = data)

# For threat outcome
threat_model <- aov(threat_score ~ ideology + intervention_factor * group_factor,
                   data = data)

# Check results for trust model
summary(trust_model)
```

```
                                  Df Sum Sq Mean Sq F value Pr(>F)
ideology                           1  115.9  115.93  74.540 <2e-16 ***
intervention_factor                2  142.7   71.35  45.880 <2e-16 ***
group_factor                       1    1.0    0.97   0.625  0.429
intervention_factor:group_factor   2    4.9    2.45   1.573  0.208
Residuals                        597  928.5    1.56
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# Check results for threat model
summary(threat_model)
```

```
                                 Df Sum Sq Mean Sq F value Pr(>F)    
ideology                          1  179.7  179.73   99.52 <2e-16 ***
intervention_factor               2  248.8  124.38   68.88 <2e-16 ***
group_factor                      1    0.6    0.60    0.33  0.566    
intervention_factor:group_factor  2    4.4    2.22    1.23  0.293    
Residuals                       597 1078.1    1.81                   
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
library(effectsize)

# For trust model
eta_squared(trust_model, partial = TRUE)
```

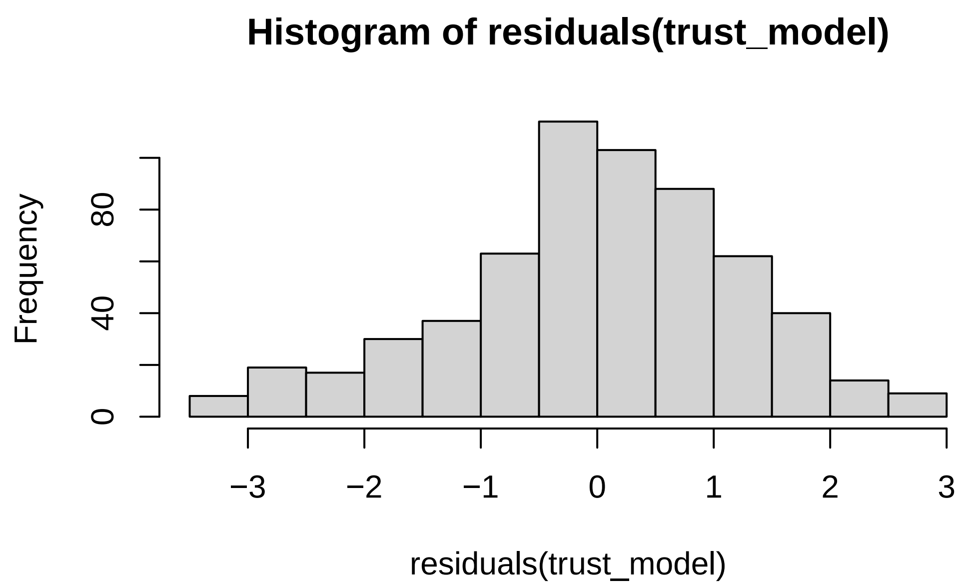
```
# Effect Size for ANOVA (Type I)

Parameter                        | Eta2 (partial) |       95% CI
-----------------------------------------------------------------
ideology                         |           0.11 | [0.07, 1.00]
intervention_factor              |           0.13 | [0.09, 1.00]
group_factor                     |       1.05e-03 | [0.00, 1.00]
intervention_factor:group_factor |       5.24e-03 | [0.00, 1.00]

- One-sided CIs: upper bound fixed at [1.00].
```

```r
# For threat model
eta_squared(threat_model, partial = TRUE)
```

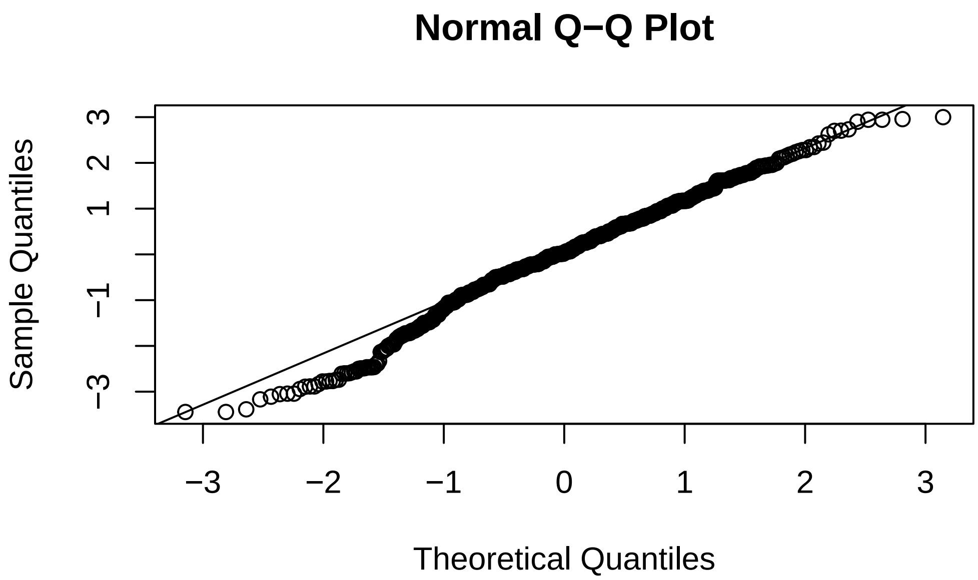
```
# Effect Size for ANOVA (Type I)

Parameter                        | Eta2 (partial) |       95% CI
-----------------------------------------------------------------
ideology                         |           0.14 | [0.10, 1.00]
intervention_factor              |           0.19 | [0.14, 1.00]
group_factor                     |       5.53e-04 | [0.00, 1.00]
intervention_factor:group_factor |       4.10e-03 | [0.00, 1.00]

- One-sided CIs: upper bound fixed at [1.00].
```

Check Model Assumptions

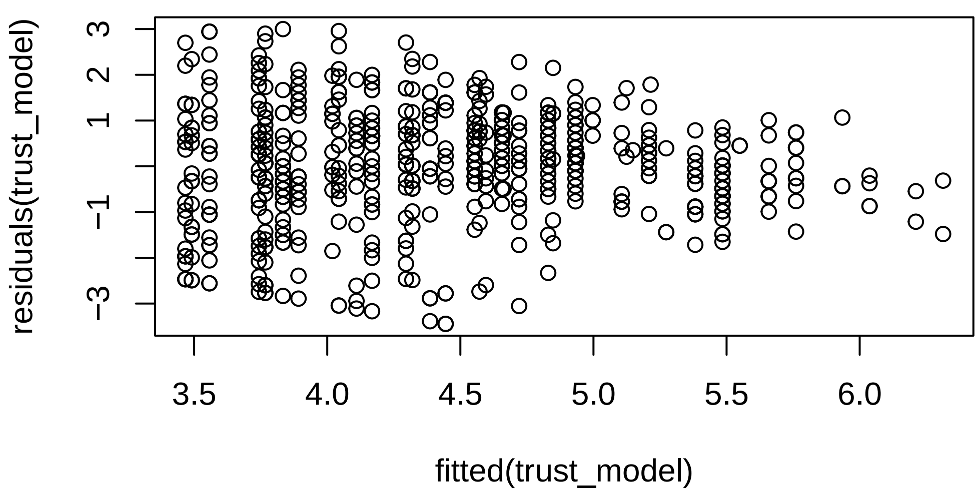
```r
# Check normality of residuals
hist(residuals(trust_model))
```




```r
qqnorm(residuals(trust_model))
qqline(residuals(trust_model))
```




```
# Check homogeneity of variance
plot(fitted(trust_model), residuals(trust_model))
```



```
# Repeat for threat model
```

Calculate EMMs at Different Levels of Perspective Taking

```r
library(emmeans)
```

```
Welcome to emmeans.
Caution: You lose important information if you filter this package's results.
See '? untidy'
```

```r
# For trust model - Examine the interaction effect
trust_emms <- emmeans(trust_model,
                     specs = ~ intervention_factor | group_factor,
                     covariates = list(ideology = mean(data$ideology, na.rm = TRUE)))

# For threat model
threat_emms <- emmeans(threat_model,
                      specs = ~ intervention_factor | group_factor,
                      covariates = list(ideology = mean(data$ideology, na.rm = TRUE)))

# Alternative: Get the full grid of means
trust_emms_grid <- emmeans(trust_model,
                          specs = ~ intervention_factor * group_factor,
                          covariates = list(ideology = mean(data$ideology, na.rm = TRUE)))

threat_emms_grid <- emmeans(threat_model,
                           specs = ~ intervention_factor * group_factor,
                           covariates = list(ideology = mean(data$ideology, na.rm = TRUE)))
```

Examine Pairwise Comparisons

```r
# Compare intervention conditions within each moral condition
trust_pairs <- pairs(trust_emms, adjust = "tukey")
summary(trust_pairs)
```

```
group_factor = Individualizing:
 contrast         estimate    SE  df t.ratio p.value
```

```
 Better - Worse     1.1890 0.171 597   6.964 <0.0001
 Better - Control   1.0986 0.178 597   6.178 <0.0001
 Worse - Control   -0.0904 0.175 597  -0.517  0.8632

group_factor = Binding:
 contrast         estimate    SE  df t.ratio p.value
 Better - Worse     1.0627 0.177 597   6.021 <0.0001
 Better - Control   0.6616 0.180 597   3.665  0.0008
 Worse - Control   -0.4011 0.177 597  -2.266  0.0615

P value adjustment: tukey method for comparing a family of 3 estimates
```

```
threat_pairs <- pairs(threat_emms, adjust = "tukey")
summary(threat_pairs)
```

```
group_factor = Individualizing:
 contrast         estimate    SE  df t.ratio p.value
 Better - Worse     -1.670 0.184 597  -9.079 <0.0001
 Better - Control   -1.314 0.192 597  -6.855 <0.0001
 Worse - Control     0.357 0.189 597   1.892  0.1417

group_factor = Binding:
 contrast         estimate    SE  df t.ratio p.value
 Better - Worse     -1.326 0.190 597  -6.970 <0.0001
 Better - Control   -0.929 0.195 597  -4.776 <0.0001
 Worse - Control     0.397 0.191 597   2.079  0.0950

P value adjustment: tukey method for comparing a family of 3 estimates
```

Moderating role of empathy

```
# Run full model with perspective taking as a continuous moderator
trust_model_full <- lm(trust_score ~ ideology + intervention_factor * group_factor * perspec
                      data = data)
summary(trust_model_full)
```

```
Call:
lm(formula = trust_score ~ ideology + intervention_factor * group_factor *
    perspective_taking, data = data)
```

```
Residuals:
    Min      1Q  Median      3Q     Max 
-3.6260 -0.6453  0.0749  0.8261  3.0567 

Coefficients:
                                                                  Estimate
(Intercept)                                                        3.52041
ideology                                                           0.29069
intervention_factorWorse                                          -1.60863
intervention_factorControl                                        -2.38973
group_factorBinding                                               -0.69687
perspective_taking                                                 0.20730
intervention_factorWorse:group_factorBinding                       1.57783
intervention_factorControl:group_factorBinding                     2.14013
intervention_factorWorse:perspective_taking                        0.10300
intervention_factorControl:perspective_taking                      0.31830
group_factorBinding:perspective_taking                             0.14606
intervention_factorWorse:group_factorBinding:perspective_taking   -0.36203
intervention_factorControl:group_factorBinding:perspective_taking -0.41751
                                                                  Std. Error
(Intercept)                                                          0.71220
ideology                                                             0.03455
intervention_factorWorse                                             1.09632
intervention_factorControl                                           1.11328
group_factorBinding                                                  1.03840
perspective_taking                                                   0.17682
intervention_factorWorse:group_factorBinding                         1.54580
intervention_factorControl:group_factorBinding                       1.55741
intervention_factorWorse:perspective_taking                          0.27323
intervention_factorControl:perspective_taking                        0.27605
group_factorBinding:perspective_taking                               0.25808
intervention_factorWorse:group_factorBinding:perspective_taking      0.38221
intervention_factorControl:group_factorBinding:perspective_taking    0.38646
                                                                  t value
(Intercept)                                                         4.943
ideology                                                            8.413
intervention_factorWorse                                           -1.467
intervention_factorControl                                         -2.147
group_factorBinding                                                -0.671
perspective_taking                                                  1.172
intervention_factorWorse:group_factorBinding                        1.021
intervention_factorControl:group_factorBinding                      1.374
```

```
intervention_factorWorse:perspective_taking                          0.377
intervention_factorControl:perspective_taking                        1.153
group_factorBinding:perspective_taking                               0.566
intervention_factorWorse:group_factorBinding:perspective_taking     -0.947
intervention_factorControl:group_factorBinding:perspective_taking   -1.080
                                                                   Pr(>|t|)
(Intercept)                                                        1.00e-
06 ***
ideology                                                           3.01e-
16 ***
intervention_factorWorse                                             0.1428
intervention_factorControl                                           0.0322 *
group_factorBinding                                                  0.5024
perspective_taking                                                   0.2415
intervention_factorWorse:group_factorBinding                         0.3078
intervention_factorControl:group_factorBinding                       0.1699
intervention_factorWorse:perspective_taking                          0.7063
intervention_factorControl:perspective_taking                        0.2494
group_factorBinding:perspective_taking                               0.5716
intervention_factorWorse:group_factorBinding:perspective_taking      0.3439
intervention_factorControl:group_factorBinding:perspective_taking    0.2804
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.238 on 591 degrees of freedom
Multiple R-squared:  0.2411,    Adjusted R-squared:  0.2257
F-statistic: 15.65 on 12 and 591 DF,  p-value: < 2.2e-16
```

```r
threat_model_full <- lm(threat_score ~ ideology + intervention_factor * group_factor * perspe
                       data = data)
summary(threat_model_full)
```

```
Call:
lm(formula = threat_score ~ ideology + intervention_factor *
    group_factor * perspective_taking, data = data)

Residuals:
    Min      1Q  Median      3Q     Max
-3.7201 -0.7292  0.0758  0.9459  3.2385

Coefficients:
```

```
                                                              Estimate
(Intercept)                                                   4.213264
ideology                                                     -0.346035
intervention_factorWorse                                      1.408156
intervention_factorControl                                    0.516718
group_factorBinding                                           0.092544
perspective_taking                                           -0.129543
intervention_factorWorse:group_factorBinding                 -0.677885
intervention_factorControl:group_factorBinding               -0.003676
intervention_factorWorse:perspective_taking                   0.067503
intervention_factorControl:perspective_taking                 0.200847
group_factorBinding:perspective_taking                        0.024135
intervention_factorWorse:group_factorBinding:perspective_taking   0.080729
intervention_factorControl:group_factorBinding:perspective_taking -0.096944
                                                              Std. Error
(Intercept)                                                     0.776600
ideology                                                        0.037678
intervention_factorWorse                                        1.195452
intervention_factorControl                                      1.213948
group_factorBinding                                             1.132295
perspective_taking                                              0.192807
intervention_factorWorse:group_factorBinding                    1.685576
intervention_factorControl:group_factorBinding                  1.698236
intervention_factorWorse:perspective_taking                     0.297940
intervention_factorControl:perspective_taking                   0.301015
group_factorBinding:perspective_taking                          0.281418
intervention_factorWorse:group_factorBinding:perspective_taking   0.416768
intervention_factorControl:group_factorBinding:perspective_taking   0.421403
                                                              t value
(Intercept)                                                     5.425
ideology                                                       -9.184
intervention_factorWorse                                        1.178
intervention_factorControl                                      0.426
group_factorBinding                                             0.082
perspective_taking                                             -0.672
intervention_factorWorse:group_factorBinding                   -0.402
intervention_factorControl:group_factorBinding                 -0.002
intervention_factorWorse:perspective_taking                     0.227
intervention_factorControl:perspective_taking                   0.667
group_factorBinding:perspective_taking                          0.086
intervention_factorWorse:group_factorBinding:perspective_taking   0.194
intervention_factorControl:group_factorBinding:perspective_taking  -0.230
                                                              Pr(>|t|)
```

```
(Intercept)                                                             8.45e-
08 ***
ideology                                                                 < 2e-
16 ***
intervention_factorWorse                                                   0.239
intervention_factorControl                                                 0.671
group_factorBinding                                                        0.935
perspective_taking                                                         0.502
intervention_factorWorse:group_factorBinding                               0.688
intervention_factorControl:group_factorBinding                             0.998
intervention_factorWorse:perspective_taking                                0.821
intervention_factorControl:perspective_taking                              0.505
group_factorBinding:perspective_taking                                     0.932
intervention_factorWorse:group_factorBinding:perspective_taking            0.846
intervention_factorControl:group_factorBinding:perspective_taking          0.818
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.35 on 591 degrees of freedom
Multiple R-squared:  0.2879,    Adjusted R-squared:  0.2735
F-statistic: 19.91 on 12 and 591 DF,  p-value: < 2.2e-16
```

```r
# Calculate EMMs at specific levels of perspective taking
pt_mean <- mean(data$perspective_taking, na.rm = TRUE)
pt_sd <- sd(data$perspective_taking, na.rm = TRUE)

trust_emms_pt <- emmeans(trust_model_full,
                        specs = ~ intervention_factor | group_factor,
                        at = list(perspective_taking = c(pt_mean - pt_sd, pt_mean, pt_mean +
                        covariates = list(ideology = mean(data$ideology, na.rm = TRUE)))
```

```
NOTE: Results may be misleading due to involvement in interactions
```

Graph

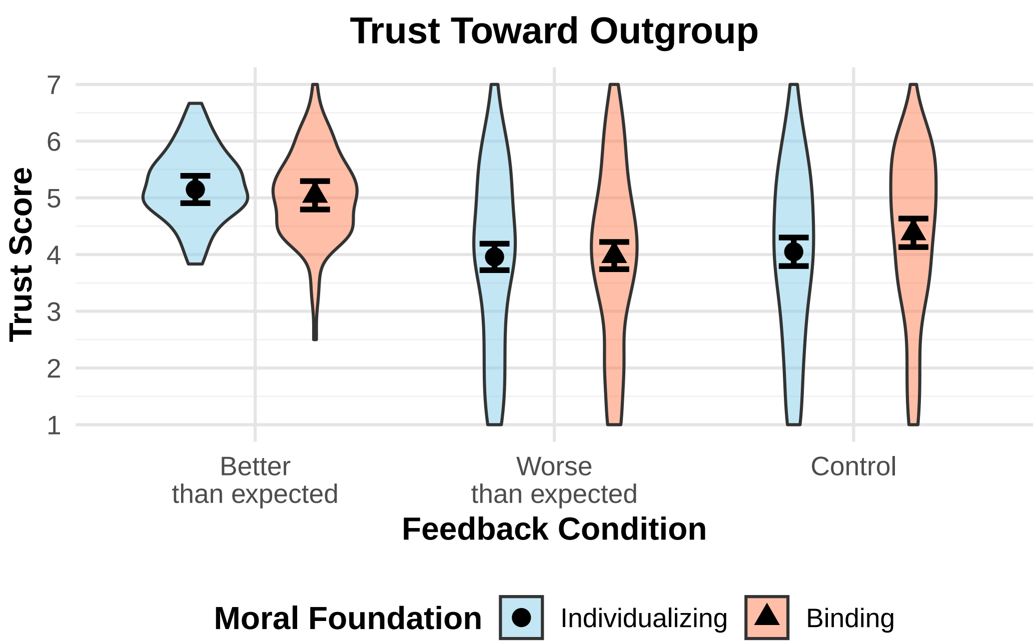
```r
library(ggplot2)
library(dplyr)
library(gridExtra)
```

```
Warning: package 'gridExtra' was built under R version 4.6.1
```

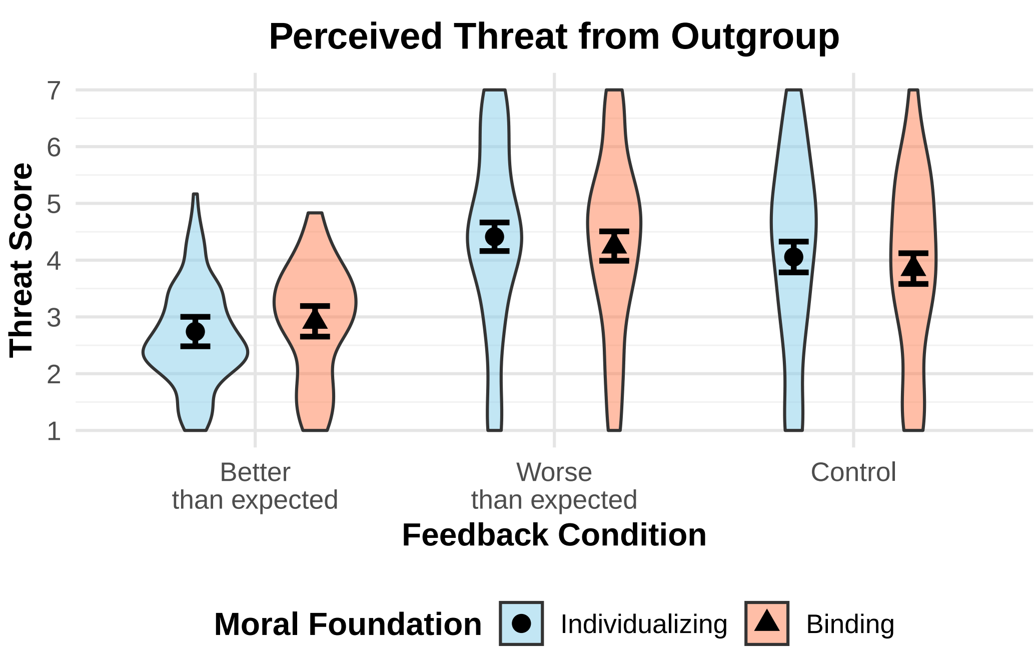
```
Attaching package: 'gridExtra'


The following object is masked from 'package:dplyr':

    combine
```

```
# First, prepare your data with EMMs
trust_emms_df <- as.data.frame(trust_emms)
threat_emms_df <- as.data.frame(threat_emms)

# Violin plot for Trust
trust_plot <- ggplot() +
  # Add violin plot of raw data
  geom_violin(data = data, aes(x = intervention_factor, y = trust_score,
                               fill = group_factor),
              alpha = 0.5, position = position_dodge(0.8), width = 0.7) +
  # Add EMM points and error bars
  geom_point(data = trust_emms_df, aes(x = intervention_factor, y = emmean,
                                       group = group_factor, shape = group_factor),
             position = position_dodge(0.8), size = 3) +
  geom_errorbar(data = trust_emms_df,
                aes(x = intervention_factor, y = emmean,
                    group = group_factor,
                    ymin = lower.CL, ymax = upper.CL),
                position = position_dodge(0.8), width = 0.2, linewidth = 1) +
  # Custom aesthetics
  scale_fill_manual(values = c("skyblue", "coral")) +
  # Set fixed y-axis range from 1 to 7
  scale_y_continuous(limits = c(1, 7), breaks = 1:7) +
  labs(title = "Trust Toward Outgroup",
       x = "Feedback Condition",
       y = "Trust Score",
       fill = "Moral Foundation",
       shape = "Moral Foundation") +
  scale_x_discrete(labels = c("Better\nthan expected", "Worse\nthan expected", "Control")) +
  theme_minimal(base_size = 12) +
  theme(
    plot.title = element_text(hjust = 0.5, face = "bold"),
    axis.title = element_text(face = "bold"),
    legend.position = "bottom",
    legend.title = element_text(face = "bold"),
```

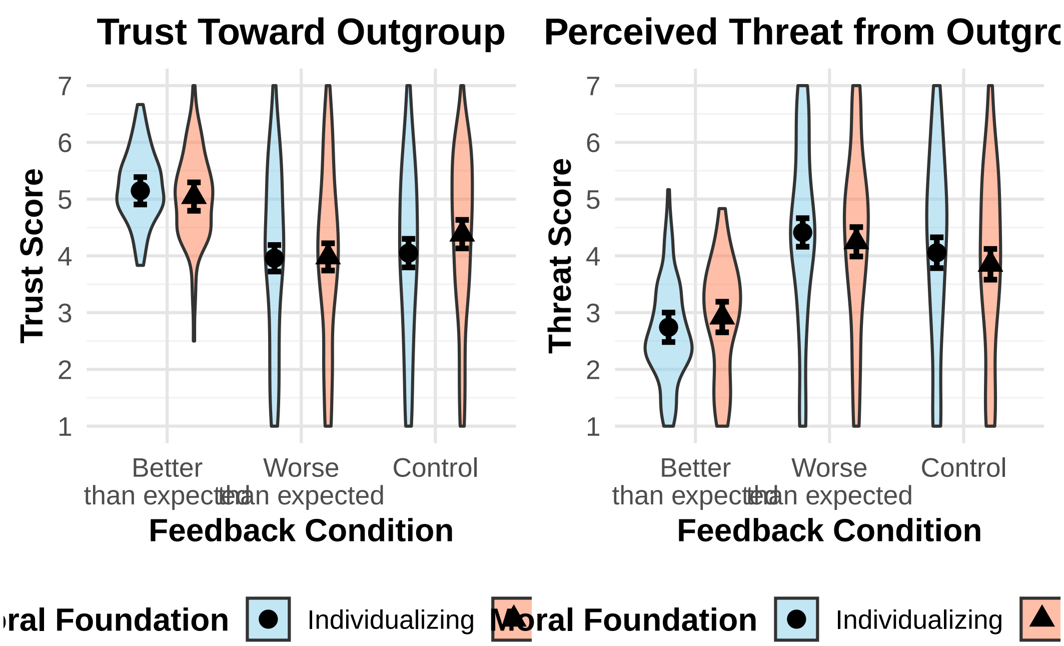
```r
    panel.grid.major = element_line(color = "gray90"),
    panel.grid.minor = element_line(color = "gray95")
  )

# Violin plot for Threat
threat_plot <- ggplot() +
  # Add violin plot of raw data
  geom_violin(data = data, aes(x = intervention_factor, y = threat_score,
                               fill = group_factor),
              alpha = 0.5, position = position_dodge(0.8), width = 0.7) +
  # Add EMM points and error bars
  geom_point(data = threat_emms_df, aes(x = intervention_factor, y = emmean,
                                        group = group_factor, shape = group_factor),
             position = position_dodge(0.8), size = 3) +
  geom_errorbar(data = threat_emms_df,
                aes(x = intervention_factor, y = emmean,
                    group = group_factor,
                    ymin = lower.CL, ymax = upper.CL),
                position = position_dodge(0.8), width = 0.2, linewidth = 1) +
  # Custom aesthetics
  scale_fill_manual(values = c("skyblue", "coral")) +
  # Set fixed y-axis range from 1 to 7
  scale_y_continuous(limits = c(1, 7), breaks = 1:7) +
  labs(title = "Perceived Threat from Outgroup",
       x = "Feedback Condition",
       y = "Threat Score",
       fill = "Moral Foundation",
       shape = "Moral Foundation") +
  scale_x_discrete(labels = c("Better\nthan expected", "Worse\nthan expected", "Control")) +
  theme_minimal(base_size = 12) +
  theme(
    plot.title = element_text(hjust = 0.5, face = "bold"),
    axis.title = element_text(face = "bold"),
    legend.position = "bottom",
    legend.title = element_text(face = "bold"),
    panel.grid.major = element_line(color = "gray90"),
    panel.grid.minor = element_line(color = "gray95")
  )

# Display both plots
trust_plot
```


Trust Toward Outgroup
7
6
5
4
3
2
1
Trust Score
Better
than expected
Worse
than expected
Control
Feedback Condition
Moral Foundation
Individualizing
Binding

```
threat_plot
```


Perceived Threat from Outgroup
7
6
5
4
3
2
1
Threat Score
Better
than expected
Worse
than expected
Control
Feedback Condition
Moral Foundation
Individualizing
Binding

```
# To save the plots to files:
ggsave("trust_violin_plot.jpg", trust_plot, width = 10, height = 7, dpi = 300)
ggsave("threat_violin_plot.jpg", threat_plot, width = 10, height = 7, dpi = 300)

# To display both plots side by side in a single figure:
combined_plot <- grid.arrange(trust_plot, threat_plot, ncol = 2)
```




```
ggsave("combined_violin_plots.jpg", combined_plot, width = 14, height = 7, dpi = 300)
```